

Project Initialization and Planning Phase

Date	05 July 2026
Team ID	
Project Name	Green-Predict-Pro
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

Green-Predict-Pro is a Smart Agricultural Production Optimization Engine that helps farmers, researchers and policymakers make data-driven decisions about which crop to cultivate. By combining soil nutrients (Nitrogen, Phosphorous, Potassium) with environmental factors (temperature, humidity, pH and rainfall), the system recommends the most suitable crop for the given conditions and highlights the productivity potential.

Project Overview

Field	Details
Objective	Provide intelligent, data-driven crop recommendations to maximize yield and resource efficiency across diverse soil and climate conditions.
Scope	Covers 20+ major crops using publicly available crop-recommendation datasets; includes ML modeling, a Flask web interface and an analytics view for research use.

Problem Statement

Field	Details
Description	Farmers often choose crops based on tradition or advice, without considering the actual soil nutrients and climate, leading to sub-optimal yields and inefficient resource use.
Impact	Better crop selection can increase yields, reduce fertilizer waste, improve farmer income and support sustainable agriculture policies.

Proposed Solution

Field	Details
Approach	Train supervised ML classifiers (Random Forest, Decision Tree, SVM, KNN) on the

	crop dataset and deploy the best-performing model behind a Flask web app.
Key Features	- ML-based crop recommendation from N, P, K, temperature, humidity, pH and rainfall. - Crop suitability & environmental assessment for a user-selected crop. - Analytical dashboard for researchers and policymakers. - Downloadable prediction report.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU / GPU specifications, number of cores	Intel Core i3 or above, 4 cores
Memory	RAM specifications	Minimum 4 GB (8 GB recommended)
Storage	Disk space for data, models, and logs	10 GB free space
Software		
Frameworks	Python web framework	Flask
Libraries	Data & ML libraries	NumPy, Pandas, scikit-learn, Matplotlib, Seaborn, SciPy
Development Environment	IDE	Jupyter Notebook, VS Code / PyCharm
Data		
Data	Source, size, format	Kaggle Crop Recommendation dataset, ~2200 rows, CSV